

Coherent Errors: Chess Transformers Play Correctly on Misremembered Boards

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Abstract

When a search-free chess transformer plays an illegal move, is it producing noise, or acting correctly on a board it has misremembered? We answer this by reading linear probes against a rules-engine oracle at every token, in models of 3,440,064 to 39,884,288 parameters. Of the illegal moves a model plays, 0.355 are legal in the position its own probe says it believes, against 0.050 for the same move scored under another position’s belief and 0.007 for an arbitrary illegal move, with a decoding ceiling of 0.654. The belief is causally load-bearing: editing the single probe direction that encodes an occupied square removes 0.934 of the probability mass on moves from it, while a norm-matched random edit removes 0.077 and shifts the mass at chance. Both effects strengthen monotonically across a twelvefold range of model size. We also find that where a world model is measured matters: whole-board probe accuracy predicts an illegal next move at AUC 0.538 to 0.671, while the same representation restricted to the squares the move uses reaches 0.592 to 0.687, the two being indistinguishable in the opening and separating with depth. Long-horizon failure here is not noise breaking through. It is competent play in a world that has drifted.

1 Introduction

A transformer stores nothing between positions. Every token’s representation is rebuilt from the whole prefix through attention (Vaswani et al., 2017), so any quantity a task requires exists only because the network re-derives it, at every layer and every position. That re-derivation is dependable over short spans and unreliable over long ones, and the usual remedies act outside the model: tree search (Yao et al., 2023), process supervision (Lightman et al., 2024), sampling with a vote (Wang et al., 2023). Each raises accuracy and none says what fails inside.

The question matters because two candidate failures recommend opposite remedies. If the model has forgotten the world, a verifier that rescores outputs pays repeatedly to repair something the architecture gave away. If the model holds the world and reasons badly over it, a verifier is exactly right. Telling them apart requires reading an intermediate representation against a known-correct intermediate state, and open-ended text has no such state.

Chess from move notation does, and that is why we use it: a rules engine gives the exact position after any prefix, legality is decidable, and depth is uniformly scaled. We are explicit that this buys measurement precision at the cost of generality, and Section 5 states what it does not license. Prior work established that such models carry board-state representations and that intervening on them changes behaviour (Li et al., 2023; Karvonen, 2024; Toshniwal et al., 2022). We take that as premise. Our question is what happens to those representations as reasoning deepens, and whether their failures explain the model’s failures.

Findings.

1. **The errors are coherent.** 0.355 of illegal moves are legal in the model’s own decoded board, against 0.050 for a mismatched belief and 0.007 for a random illegal move.
2. **The belief is read, not merely present.** A calibrated edit to one probe direction removes 0.934 of the mass on moves from the affected square; a norm-matched random edit removes 0.077.
3. **Where the world model is measured matters, increasingly with depth.** Aggregate probe accuracy predicts illegality at AUC 0.538 to 0.671; restricted to the squares the move uses, 0.592 to 0.687.

Table 1: State decodability against every trivial explanation, for the baseline model. Occupied-square accuracy at the most decodable layer, beside a per-square per-depth majority predictor, a label-permutation control, and a randomised-weight model of identical architecture. The gap is what training put there.

Ply	model	majority	label perm.	random weights	illegal rate
0–10	0.975	0.867	0.879	0.898	0.0004
10–20	0.876	0.599	0.673	0.641	0.0115
20–30	0.761	0.444	0.487	0.439	0.0439
30–40	0.655	0.359	0.315	0.297	0.0982
40–50	0.557	0.261	0.166	0.195	0.1766
50–60	0.475	0.073	0.072	0.141	0.2347
60–80	0.353	0.000	0.019	0.071	0.2920
80–120	0.173	0.000	0.001	0.038	0.3231
120–161	0.066	0.000	0.000	0.030	0.3192

2 Setup

Data and oracle. 120,000 human games above a rating floor, deduplicated and split *by game* so no prefix is shared between probe fitting and evaluation: 96,000 for training, 12,000 for probes, 12,000 for evaluation, mean length 78.2 plies. A rules engine emits the exact state after every token. Models are decoder-only, trained on move notation alone, with no search at inference.

Probes and controls. One linear head per layer predicting per-square occupancy, fitted on the probe split and reported on the evaluation split. Because most squares are empty, we report accuracy on occupied squares against three controls: a per-square per-depth majority predictor, a label-permutation control that pairs activations with another game’s state at equal depth, and a randomised-weight model of identical architecture (Table 1, following Hewitt and Liang, 2019).

Observables. State-tracking error is observed behaviourally as the rate at which the model’s top-1 move is illegal in the true position, and representationally as probe-oracle disagreement.

3 Results

3.1 State and behaviour decay together

Occupied-square decodability falls from 0.975 in the opening to 0.066 in the deepest bucket, while the illegal top-1 rate rises from 0.0004 to 0.319 and the mass on legal moves falls from 0.993 to 0.455 (Figure 1). At mid-game the trained model is clear of every trivial explanation, 0.555 against 0.261 for the majority predictor. There is a real state representation, and it is being lost.

3.2 Where the world model is measured matters

Two curves falling together is not evidence that one causes the other. We score aggregate and action-relevant state error identically, as predictors of the same binary outcome on the same samples, within depth so neither can simply track depth (Table 2).

Aggregate board fidelity predicts an illegal move at AUC 0.538 to 0.671, which is above chance and not negligible. The same representation restricted to the two squares the move touches reaches 0.592 to 0.687. In the opening the two are nearly indistinguishable; the gap opens with depth, and in a joint fit the action-relevant term carries a standardised weight of 0.504 against 0.118. Whole-board probe accuracy, the quantity usually reported as evidence of a world model, is a usable but weak proxy for whether that model is load-bearing, and it weakens exactly where long-horizon behaviour matters.

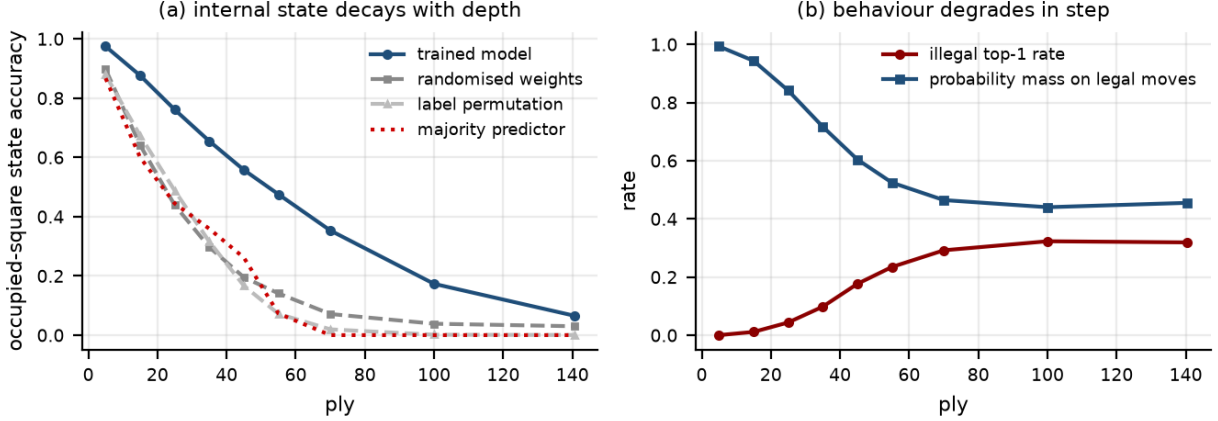


Figure 1: Internal state and external behaviour decay together. (a) Occupied-square accuracy against three controls. (b) Illegal top-1 rate and the probability mass on legal continuations.

Table 2: Aggregate and move-touched state error compared like for like, as predictors of the same binary outcome on the same samples: whether the model’s top-1 move is illegal. AUC is computed within each depth bucket so neither predictor can simply be tracking depth. The standardised coefficients come from a logistic fit containing both predictors. The aggregate measure is informative and it is the weaker of the two, and the gap widens with depth.

Ply	AUC aggregate	AUC move-touched	β aggregate	β move-touched	n
10–20	0.671	0.687	+0.065	+0.153	15,000
20–30	0.627	0.681	+0.120	+0.332	15,000
30–40	0.584	0.662	+0.100	+0.506	14,511
40–50	0.548	0.643	+0.057	+0.573	13,176
50–60	0.551	0.617	+0.114	+0.500	11,262
60–80	0.538	0.592	+0.076	+0.431	16,426
pooled, depth as covariate	n/a	n/a	+0.118	+0.504	85,375

3.3 The errors are coherent

If state loss drives the failure, an invalid action should be valid in the state the model believes it is in. We reconstruct that believed position from the probe, clearing castling rights so nothing is licensed by an assumption we did not decode, and test pseudo-legality because a decoded board may lack a king.

Of the illegal moves the model plays, 0.355 are legal in its own believed position, with a bootstrap interval of $[0.346, 0.362]$. Two controls bound the alternatives. The same move scored against a believed position decoded elsewhere at equal depth is legal 0.050 of the time, so the effect is specific to this belief at this position rather than to plausible boards generally. An arbitrary illegal move scored against the model’s own believed position is legal 0.007 of the time, so the believed board is not permissive. The ceiling on what this test can detect is 0.654, the rate at which genuinely legal moves are recovered, which is below one because probe decoding is lossy.

3.4 Inducing a false belief changes the action

The result above is observational. We therefore intervene: at a position where the side to move genuinely occupies a square, we push the residual stream along the probe direction encoding “this square is empty” and ask whether the model stops proposing moves from it.

Edit strength must be calibrated rather than chosen, because a large enough push degrades the forward pass and moves behaviour for reasons unrelated to belief. Below a quarter of the residual-stream magnitude

Table 3: Does state loss explain behaviour? The aggregate column is the within-depth AUC of whole-board probe error as a predictor of an illegal move, reported as a range over depth buckets. Move-touched error is the probability the probe is wrong on the origin or destination square. Belief consistency is the share of illegal moves admissible under the model’s own decoded board, beside the same move under another position’s belief, an arbitrary illegal move, and the legal-move recovery reference.

Model	Params	aggregate AUC	move-touched		belief consistency			
			illegal	legal	own	mismatch	random	recovery
Attention-only	7,345,920	0.54 to 0.67	0.806	0.384	0.355	0.050	0.007	0.654
Attention-only, parameter-matched	7,505,976	n/a	0.804	0.391	0.350	0.051	0.007	0.650
Recurrent channel ($\lambda=0$)	7,508,416	n/a	0.801	0.384	0.358	0.053	0.007	0.658
Recurrent channel ($\lambda=1$)	7,508,416	n/a	0.806	0.399	0.366	0.093	0.021	0.656
6L/192	3,440,064	0.52 to 0.70	0.831	0.447	0.266	0.049	0.006	0.581
12L/384	22,835,328	0.53 to 0.71	0.747	0.327	0.412	0.056	0.009	0.713
12L/512	39,884,288	0.53 to 0.74	0.711	0.297	0.439	0.059	0.009	0.745

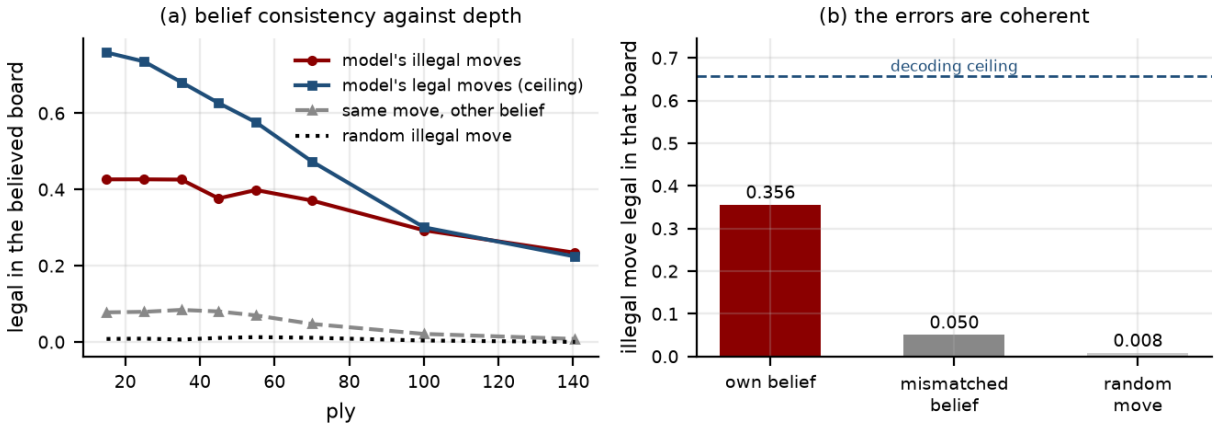


Figure 2: (a) Belief consistency against depth, restricted to buckets with at least thirty illegal cases. (b) Pooled, with the decoding ceiling marked.

the edit does not reliably flip the decoded belief; above roughly one, the norm-matched random control also destroys the output distribution. We report at 0.50, inside that window (Figure 3).

At that strength the manipulation check succeeds in 1.000 of cases, the state edit removes 0.934 of the probability mass on moves from the affected square, and the norm-matched random edit removes 0.077 while changing the mass in 0.484 of cases, which is a coin flip. The state representation is read.

3.5 Both effects strengthen with scale

Across six rungs from 3,440,064 to 39,884,288 parameters, three seeds each, belief consistency normalised by each model’s own decoding ceiling rises monotonically from 0.476 to 0.587, while the mismatched control stays flat (Figure 4). The mechanism is not an artifact of the smallest models: larger models make a *larger* fraction of their errors coherently.

A horizon that does not depend on behaviour. Behavioural horizons are fragile: ablating the recurrent read-out of a variant model *lowers* its illegal-move rate while *raising* its language-model loss, because the ablated model hedges toward the legal-move prior (Appendix). We therefore also define H_{fid} , the depth at which the probe first fails on the squares the chosen move touches more than half the time. It never consults the output distribution, so hedging buys nothing. It runs from 33.8 to 59.1 ply across the ladder against 21.4 to 33.8 for the behavioural horizon, ordering the models identically.

Table 4: Causal intervention at the calibrated edit strength. Editing the state direction removes probability mass from moves out of the affected square; a norm-matched random direction does not. The manipulation check reports how often the edit actually changed the decoded belief.

Model	n	belief flipped	mass before	mass after	random edit after
Attention-only	700	1.000	0.071	0.005	0.065
Attention-only, parameter-matched	600	1.000	0.074	0.004	0.066
Recurrent channel ($\lambda=0$)	600	1.000	0.070	0.004	0.064
Recurrent channel ($\lambda=1$)	600	1.000	0.067	0.004	0.062

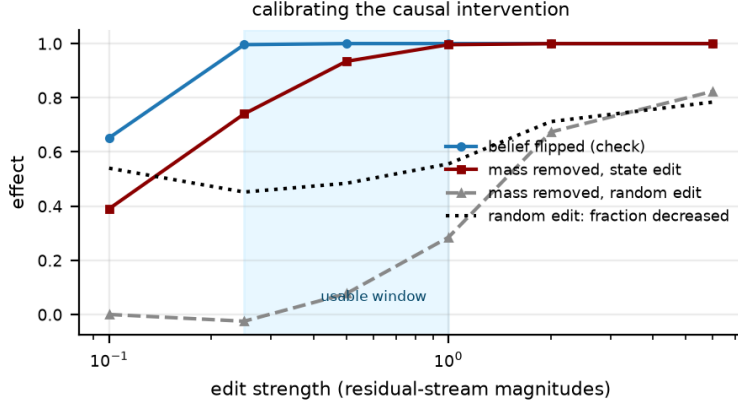


Figure 3: Calibrating the intervention. Too small and the belief does not flip; too large and a random direction also destroys the output. We report inside the shaded window.

4 What failed

Four results went against us and are reported where they changed a decision. We predicted state would be most decodable mid-stack; it is maximal at the final block, rising from 0.323 to 0.543. Our pre-registered horizon, defined on exact-position fidelity, is degenerate: that quantity is zero past ply 0 in every condition. An auxiliary state objective raises probe accuracy to the level of a model twice the size, 0.632 against 0.632, while buying a small fraction of its behavioural gain. And a second oracle-bearing domain took seven designs to reach a measurable regime, then failed to replicate for its attention-only condition, because the task permits resolving one queried object on demand rather than maintaining state. All four are detailed in the appendix.

5 Discussion and limitations

The practical content is a change in what to measure. Reporting that a probe recovers a world model at some aggregate accuracy is weak evidence that the model is load-bearing; the same representation scored on the state an action consumes is substantially more informative, increasingly so with depth. Any team instrumenting a system for long-horizon reliability should condition on the state the action uses.

The belief-consistency result also reframes the failure. A model that plays a coherent move in a misremembered position is not behaving erratically, and treatments assuming erratic behaviour aim at the wrong target: sampling several chains and voting will not help if every chain is drawn from the same drifted state.

What this does not show. Our models span 3,440,064 to 39,884,288 parameters in a single domain. We make no claim about frontier-scale systems or about natural language, and the natural-language transfer test is absent rather than approximated. Probe decoding is lossy, which is why the ceiling is reported rather than assumed; belief consistency is tested with pseudo-legal moves, which the random-move control is designed

Table 5: The scale ladder, three seeds per rung. H_{ill} is read off behaviour, the ply at which the illegal-move rate crosses five percent. H_{fid} is read off the representation, the depth at which the probe first fails on the squares the chosen move touches more than half the time, and so cannot be moved by hedging. Belief consistency is normalised by each model’s own legal-move recovery reference, because that reference rises with size. Mean $\pm$ s.d. over seeds.

Model	Params	seeds	$H_{\text{ill}}(.05)$	$H_{\text{fid}}(.5)$	belief / reference	mismatch
6L/192	3,440,064	3	21.4 ± 0.1	33.8 ± 0.2	0.476 ± 0.017	0.051 ± 0.002
8L/256	7,345,920	3	26.1 ± 0.0	40.8 ± 0.5	0.534 ± 0.008	0.052 ± 0.002
12L/256	10,504,960	3	27.8 ± 0.2	42.1 ± 0.3	0.552 ± 0.007	0.051 ± 0.002
8L/384	15,737,472	3	29.5 ± 0.2	50.0 ± 0.4	0.568 ± 0.008	0.053 ± 0.001
12L/384	22,835,328	3	31.7 ± 0.3	52.5 ± 0.5	0.577 ± 0.005	0.053 ± 0.003
12L/512	39,884,288	3	33.8 ± 0.1	59.1 ± 0.5	0.587 ± 0.011	0.056 ± 0.004

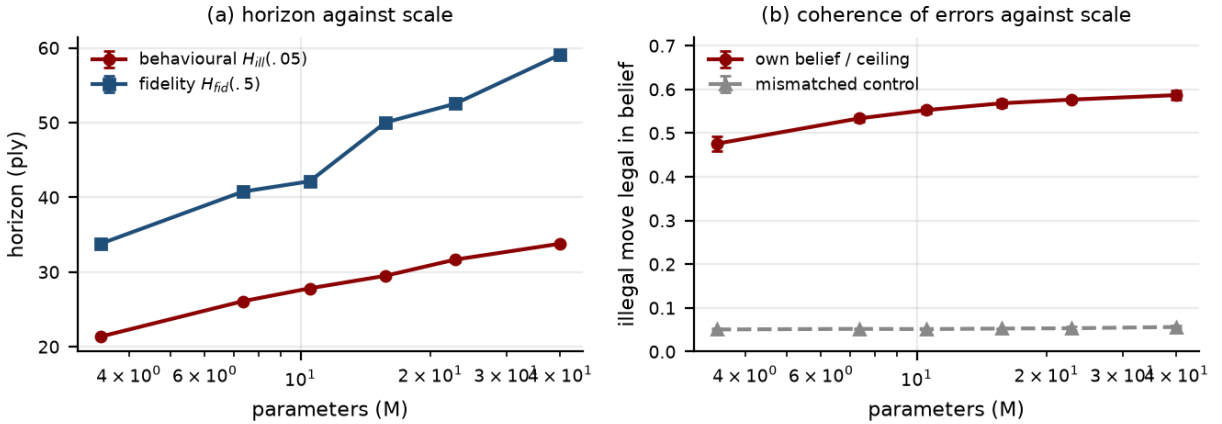


Figure 4: (a) Both horizons against scale. (b) Belief consistency normalised by each model’s decoding ceiling, with its matched control. Three seeds per rung.

to bound. The strongest available next test is a domain with an exact oracle that is neither chess nor constructed as a benchmark, and we argue program execution traces are that domain.

6 Conclusion

Measured against an exact oracle, long-horizon failure in these models separates into memory and judgement, and the memory component behaves in a specific and somewhat unsettling way. The model does not degrade into noise. It plays a reasonable move in a board it has misremembered, and it does so more consistently as it gets larger.

A The intervention we evaluated with the diagnostic

To test whether the diagnostic discriminates between architectures, we trained a variant with a fixed-width gated recurrent channel inserted after every k -th attention block, carrying $z_t \in \mathbb{R}^{d_s}$ with $d_s \ll d$, read back into the residual stream, with an optional auxiliary loss supervising z_t to be the state. Architecture, supervision and read-out are varied independently against a parameter-matched attention-only control.

With the auxiliary objective active, state decodability rises sharply, from 0.555 to 0.632 at matched parameters and tokens. The architecture without the objective, 0.556, is indistinguishable from baseline. Behaviour barely follows: the illegal rate improves from 0.176 to 0.166 and the horizon from 26.1 to 26.5 ply, while a plain attention model of twice the size reaches almost the same decodability, 0.632, and converts it

Table 6: Horizon and state fidelity by condition. $H_{\text{ill}}(.05)$ is the ply at which the illegal-move rate first crosses five percent, interpolated between bucket midpoints because the bucketed read-off returns the same bucket for every condition and so discriminates nothing. Mean $\pm$ s.d. over seeds where more than one was run.

Model	Params	seeds	occ. acc. @40–50	illegal @40–50	$H_{\text{ill}}(.05)$
Attention-only	7,345,920	3	0.555 ± 0.003	0.176 ± 0.002	26.1 ± 0.0
Attention-only, parameter-matched	7,505,976	3	0.554 ± 0.002	0.172 ± 0.003	26.0 ± 0.1
Recurrent channel ($\lambda=0$)	7,508,416	3	0.556 ± 0.003	0.169 ± 0.002	26.2 ± 0.3
Recurrent channel ($\lambda=1$)	7,508,416	3	0.632 ± 0.003	0.166 ± 0.002	26.5 ± 0.2
read-out ablated at test	n/a	3	0.559 ± 0.005	0.149 ± 0.005	27.2 ± 0.4
6L/192	3,440,064	3	0.468 ± 0.006	0.245 ± 0.003	21.4 ± 0.1
8L/256	7,345,920	3	0.555 ± 0.003	0.176 ± 0.002	26.1 ± 0.0
12L/256	10,504,960	3	0.577 ± 0.005	0.149 ± 0.001	27.8 ± 0.2
8L/384	15,737,472	3	0.632 ± 0.006	0.134 ± 0.004	29.5 ± 0.2
12L/384	22,835,328	3	0.656 ± 0.001	0.112 ± 0.002	31.7 ± 0.3
12L/512	39,884,288	3	0.694 ± 0.003	0.101 ± 0.001	33.8 ± 0.1

Table 7: Planning error against how strictly we condition on state. Each cell gives the median engine centipawn loss, the blunder rate, and the sample size, over legal moves whose touched squares the probe recovers and with at most the stated number of other squares wrong. Tightening the conditioning lowers the loss, so part of what looks like degraded judgement is state loss elsewhere on the board. Loss still grows with depth at fixed conditioning, so part of it is not. Cells with fewer than thirty scored positions are reported as n/a, and that scarcity is itself the finding: at depth a nearly intact board is rare.

Ply	≤ 0	≤ 2	≤ 4	≤ 8	≤ 16	any
20–30	n/a	24 / 0.18 (115)	24 / 0.21 (300)	31 / 0.22 (300)	44 / 0.31 (300)	58 / 0.34 (300)
30–40	n/a	n/a	38 / 0.25 (85)	62 / 0.36 (300)	75 / 0.41 (300)	91 / 0.47 (300)
40–50	n/a	n/a	n/a	67 / 0.41 (300)	120 / 0.53 (300)	111 / 0.52 (300)

into a much larger behavioural gain. An auxiliary loss that makes state linearly available does what it is asked; being available is not being used.

An ablation that improves behaviour. Zeroing the read-out at test time lowers the illegal rate, from 0.166 to 0.149, and lengthens the horizon. Language-model loss rises when it is removed, from 3.010 to 3.199 across seeds, and the read-out injects a vector of 0.150 of the residual stream’s norm, so it is not vestigial. The ablated model is worse at prediction and better at staying legal because it places more mass on legal moves, 0.647 against 0.615, while committing less to any one. The illegal rate is therefore not a monotone proxy for state quality, which is why we also report H_{fid} .

B How much of the planning error is really memory

The conditioning that would isolate judgement cleanly is unattainable: requiring the probe to recover the exact board leaves no samples past the opening. Requiring only the squares the move touches leaves samples everywhere but does not constrain the rest of the board. We therefore vary the strictness.

Tightening lowers the loss substantially: at ply 30 to 40 the median centipawn loss falls from 91 with the board unconstrained to 38 when at most four other squares may be wrong, and the blunder rate from 0.473 to 0.247. Roughly half of what a naive reading calls degraded judgement is misattributed memory failure. It does not go to zero: at fixed conditioning the loss still grows with depth, from 24 to 38. Memory is the dominant component and not the only one.

Table 8: Domain 2, permutation tracking. Answer accuracy at the shallowest and deepest depths against a chance rate of one in four, the probe’s recovery of the queried object at depth, and belief consistency beside its mismatched control. The attention-only model solves the task well above chance while its state stays at chance under the probe, so it is not maintaining a linearly decodable assignment. The supervised channel is, and its errors are coherent with it.

Model	seeds	acc. shallow	acc. deep	probe deep	belief / control
Attention-only	3	1.000 ± 0.000	0.589 ± 0.010	0.263 ± 0.010	0.256 ± 0.005 / 0.249 ± 0.003
Recurrent channel ($\lambda=1$)	3	1.000 ± 0.000	0.989 ± 0.001	0.983 ± 0.001	0.701 ± 0.017 / 0.254 ± 0.009

C A second domain, and why it did not replicate

We attempted a second oracle-bearing domain. Four arithmetic designs floored because they required modular arithmetic before state tracking became visible; training loss sat at the generator’s token-type entropy, the value a model reaches by learning the format and nothing else. A fifth, permutation tracking over five objects, also floored, and instructively: with five objects the state is an element of S_5 , the smallest non-solvable symmetric group, which is precisely the case bounded-depth models are known to be unable to track (Merrill et al., 2024). Four objects give the solvable S_4 . A sixth was all-or-nothing, admitting no partial credit; querying after every operation supplies it.

The seventh design is measurable, and the outcome is not a replication. With the supervised channel the task is nearly solved at every depth and belief consistency reproduces, 0.701 against a mismatched control of 0.254. The attention-only model solves the task well above chance, 0.589 at depth, while its probe recovers state at 0.263, which is chance. The same probe procedure recovers 0.983 in the other model, so this is a fact about the model. The task permits resolving one queried object on demand by tracing its history, so it never obliges the model to maintain an assignment. Chess allows no such shortcut. A task can carry an exact oracle, be learnable, and still fail to measure state maintenance.

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